HFP Statement of Work

Title: Quadrupole-Orbitrap Hybrid Mass Spectrometer

I. Purpose

The Human Foods Program, Office of Laboratory Operations and Applied Science, Office of Toxicology, Division of Analytical Chemistry, Spectroscopy Mass Spectrometry Branch (HFP/OLOAS/OCT/DAC/SMSB) requires the purchase of a quadrupole-orbitrap hybrid mass spectrometer for the development of qualitative and quantitative proteomic research methods. The high mass resolution, increased sensitivity, and high scan speeds of a quadrupole-orbitrap hybrid mass spectrometer provide the analytical capabilities required for the analysis of complex mixtures of proteins in FDA regulated foods and human cell samples exposed to potentially toxic chemical compounds found in human foods.

II. Background

OLOAS/OCT/DAC/SMSB requires the purchase of a quadrupole-orbitrap hybrid mass spectrometer for use in the development and implementation of qualitative and quantitative mass spectrometry-based proteomics methods. These methods will be used for, but not limited to, the detection and quantitation of 1) proteins found in FDA regulated foods, and 2) proteins isolated from human cell lines following exposure to potential toxic compounds found in human foods. Such methods will allow for the detection and quantitation of allergenic proteins, protein toxins, transgenic proteins, bacterial or viral proteins, and emerging alternative proteins found in the human food supply. Furthermore, these methods will also be used to support our Division of Toxicology's goals of developing New Approach Methods (NAMs) to assess the toxicological effects of various chemical constituents found in botanical dietary supplements and human foods regulated by the FDA. Each of these proteomics applications requires the analysis of different sample matrices containing a complex mixture of proteins covering a wide dynamic range of concentrations. A quadrupole-orbitrap mass spectrometer is needed to achieve the required level of acquisition speed (70 Hz), sensitivity (<1.5 fg), high mass accuracy (<1 ppm) and high resolving power (480,000 at *m/z* 200) needed to overcome this major analytical challenge. In addition, a quadrupole-orbitrap mass spectrometer provides the operational flexibility of acquiring data in different modes including non-targeted, data-independent or data-dependent acquisition (DIA or DDA), targeted, parallel reaction monitoring (PRM), or hybrid DIA-PRM modes, depending on the application.

III. Objectives

The objective of this requirement is to purchase a quadrupole-orbitrap mass spectrometer, which is critical to support the Agency's mission of developing proteomics methods for the detection and quantitation of proteins in FDA regulated food products and to provide methods to support the development of NAMs methods to assess the potential toxicity of a variety of chemical compounds found in human foods.

The major components required for this quadrupole-orbitrap mass spectrometer include the following:

- Ion source housing with automatically connected gas and voltage connections. Ion source must include one HESI spray insert and a calibration probe.

- Ion Source which generates internal calibrant ions for real-time mass calibration of any mass spectrum

- Independent syringe pump controlled via the data station for maximum flexibility.

- Vacuum system pumped by a 6-stage split-flow turbomolecular pump and one single stage turbo pump

- High-capacity transfer tube and an electrodynamic ion funnel to capture and efficiently focus ions

- Resolving injection filter and bent trap to improve robustness and to reduce noise by preventing neutral and high-velocity clusters from entering the quadrupole mass filter

- Segmented quadrupole mass filter with rod-switching capabilities in front of the curved ion trap (C-Trap) with a hyperbolic surface

- High-field orbitrap mass analyzer with enhanced FT transient processing

- Ion routing multipole for trapping of ions and higher-energy collisional dissociation

- Integrated computer-based data system capable of controlling the instrument and associated, user selected, liquid chromatograph system, acquiring data, and performing subsequent data analysis. The instrument shall be supplied with appropriate data analysis software.

- Maintenance Consumables Kit- Consumables required to perform the maintenance of the quadrupole and bent flatapole

- Factory Warranty PM - Annual standard preventive maintenance visit and a standard preventive maintenance Kit
- The vendor shall provide an option for an uninterruptable power supply for system operation.
- The instrument purchased shall include the following additional considerations:
    a) The instrument shall have an integrated computer-based data system capable of controlling the instrument and associated, user selected, liquid chromatograph system, acquiring data, and performing subsequent data analysis. The instrument shall be supplied with appropriate data analysis software.
    b) The instrument shall be sold with a service contract to cover parts and labor for one (1) year after instrument installation.
    c) The vendor shall include a personalized training program. Training will be web-based and instructor-led at the CFSAN location.
    d) The vendor shall supply an uninterruptable power supply for system operation.

IV. Requirements

The Contractor shall provide an instrument that meets the technical requirements listed below.

1. The ion source of the instrument must provide the following:

- Enhanced heat transfer for electrospray stability across a broad temperature range
- Adjustable and reproducible movement of the probe in the x, y, and z positions
- Versatile flow rate compatibility—supports a wide range of flow rates from 1 to 2,000 µL/min, providing flexibility for various applications
- Compatibility with an Atmospheric Pressure Chemical Ionization (APCI) probe with flow rates from 100 to 2,000 µL/min and can be upgraded with Atmospheric Pressure Photoionization (APPI) capabilities for expanded functionality
- Compatibility with an existing nano-spray ion (Thermo Scientific™ NanoSpray Flex NG™) which enables high nano-ESI flexibility and probe position adjustment

2. The ion optics of the instrument must provide the following:

- A high-capacity ion transfer tube (HCTT) for increased ion flux into the vacuum system
- An Electrodynamic Ion Funnel (EDIF) to efficiently capture ions as they leave the HCTT

- A bent trap to reduce noise to prevent neutrals and high-velocity clusters from entering the quadrupole mass filter using a double-bent design geometry

- A bent trap that can be operated in trapping mode for operation at high instrument duty cycles

3. The mass analyzers of the instrument must provide the following:

- A segmented quadrupole mass filter with rod-switching capabilities

- A segmented quadrupole mass filter for precursor ion selection with a variable precursor isolation width from *m/z* 0.4 to *m/z* 2,000 and MS/MS precursor ion selection with high transmission from *m/z* 40 to 2,500.

- An ion-routing multipole for ion trapping during MS scans and HCD

- Automatic gain control capabilities for controlled injection of ions

- A high-field orbitrap mass analyzer with 4kV electrode voltage

- A high-field orbitrap mass analyzer with enhanced FT transient processing resulting in resolving power from 3,750 up to 480,000 FWHM at *m/z* 200 and isotopic fidelity

- A high-field orbitrap mass analyzer with up to 70 Hz acquisition speed

- Enhanced Dynamic Range (eDR) mode enabling up to 5 orders of magnitude of intra-scan dynamic range.

- Ability to operate in data-dependent acquisition (DDA), data-independent acquisition (DIA), parallel reaction monitoring (PRM), and hybrid PRM-DIA modes.

- A TurboTMT mode ions are detected with Phi-SDM processing.

4. *The* ***vacuum system of the instrument must provide the following:***

- A compact, single turbomolecular pump design that regulates the adequate vacuum in six stages for the aluminum high-vacuum analyzer chambers

- Advanced vacuum technology that reduces pressure in the ultrahigh vacuum regions, enhancing the transmission of ions to the orbitrap mass analyzer.

5. ***The data acquisition system of the instrument must provide the following:***

- The instrument shall have an integrated data system capable of calibrating and controlling the instrument and associated liquid chromatographs, performing diagnostics, acquiring data, and performing subsequent data analysis.

- The instrument shall be supplied with appropriate data analysis software.

- A method editor application with a comprehensive application specific template library, method setup supported by tooltips and a drag-and-drop user interface to facilitate method DDA, DIA, Hybrid-DIA method development.

6. ***Additional Vendor requirements***

- The system shall be a newly manufactured unit, not used or refurbished, or previously used for demonstration.
- The vendor shall ship all parts and equipment
- The vendor shall deliver and install the instrument. The HFP project officer will perform inspection and acceptance of the mass spectrometer system.
- The vendor shall demonstrate upon installation that the item will meet all performance specifications by the manufacturer.
- The instrument shall be sold with a service contract to cover parts and labor for one (1) year after instrument installation.
- The vendor shall include a personalized training program. Training will be instructor-led at the HFP location.
- The vendor shall supply an uninterruptable power supply for system operation.
- The vendor shall provide trade-in service/deinstallation labor on existing instrumentation

V. Deliverables

The vendor shall deliver and install the instrument and all components specified above to the FDA/HFP laboratory facility located at 5001 Campus Drive, College Park, MD 20740. Delivery shall be made prior to September 30, 2026. If the vendor is not performing the installation on the day of delivery, the delivery shall be made to the loading dock and moved into location on the date of installation. The vendor shall demonstrate upon installation that the item will meet all performance specifications by the manufacturer. The instrument shall not be accepted until those performance specifications have been met. The HFP project officer will perform inspection and acceptance of the quadrupole-orbitrap mass spectrometer. The entire system shall be warranted for parts and labor for twelve (12) months from date of installation. Service and installation shall be provided by service engineers who are trained and certified by the original manufacturer of the instrument. Engineers should have access to the manufacturer's latest technical developments, repair procedures, application updates, diagnostic software, and planned maintenance procedures. A three (3) day on-site installation and training for FDA users will be arranged by the HFP project officer.

VI. Government-Furnished Property, material, Equipment, or Information (GFP, GFM, GFE, or GFI)

N/A

VII. Contractor Provided Special Materials Requirements

N/A

VIII. Security

N/A

IX. Travel

N/A

X. Section 508 Compliance

N/A

XI. Other Considerations

N/A

XII. Place of Performance

The work shall be performed at FDA/HFP, 5001 Campus Drive, College Park, Maryland, 20740. Delivery and installation of the instrument and all components specified above shall be coordinated through the HFP Project Officer. Work shall be conducted Monday through Friday, excluding federal holidays.

XIII. Period of Performance

Delivery shall be made prior to September 30, 2026. The entire system shall be under warranty for parts and labor for twelve (12) months from date of installation.